

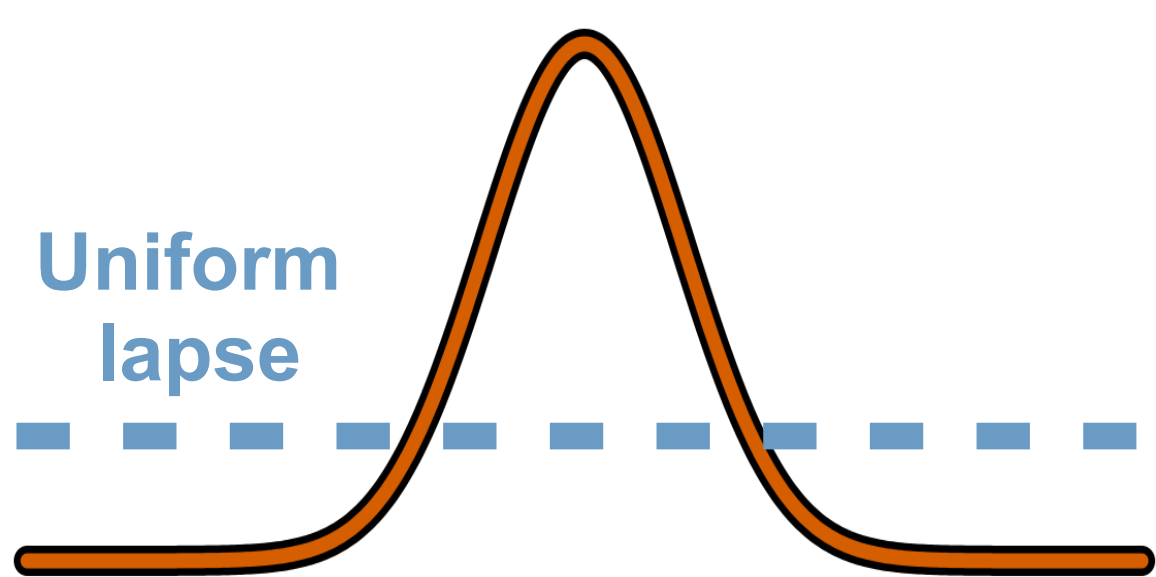
Guessing reveals internal models of perceptual precision

Caroline Myers, Chaz Firestone, and Justin Halberda, Johns Hopkins University

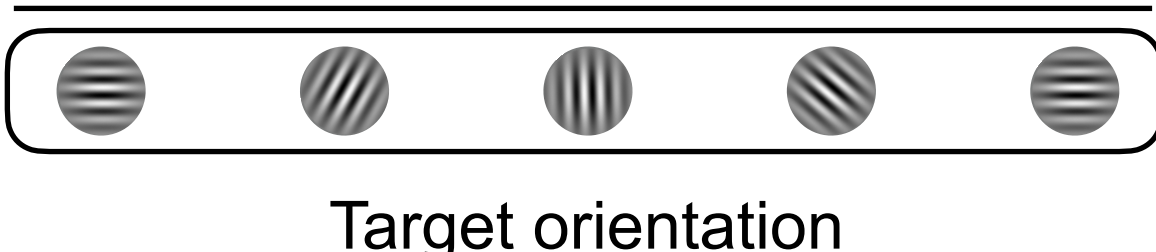
Significance

Guessing is regarded as **theoretically uninformative**...

Internal representation

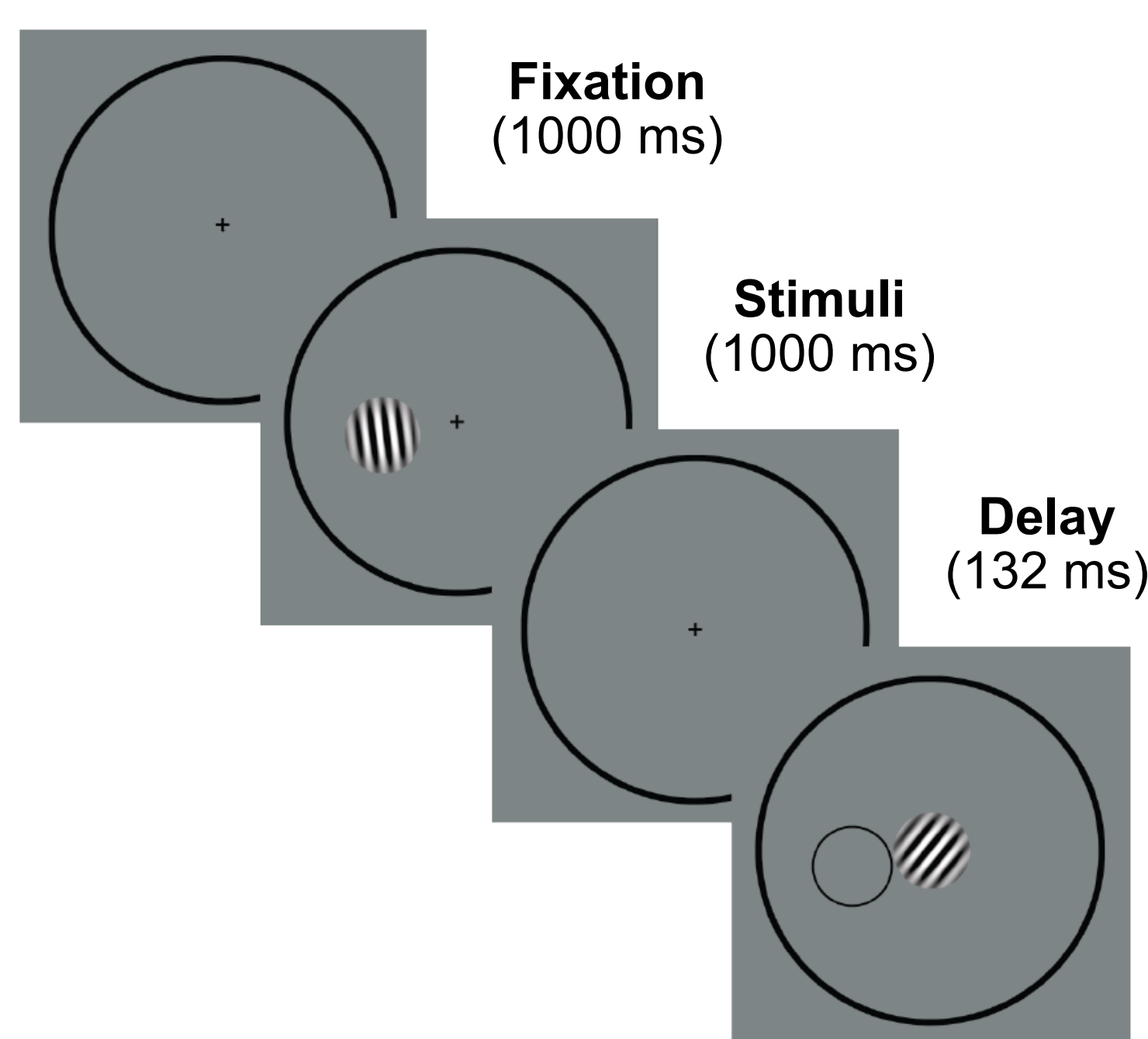


Might guessing be systematically **structured** in feature space?

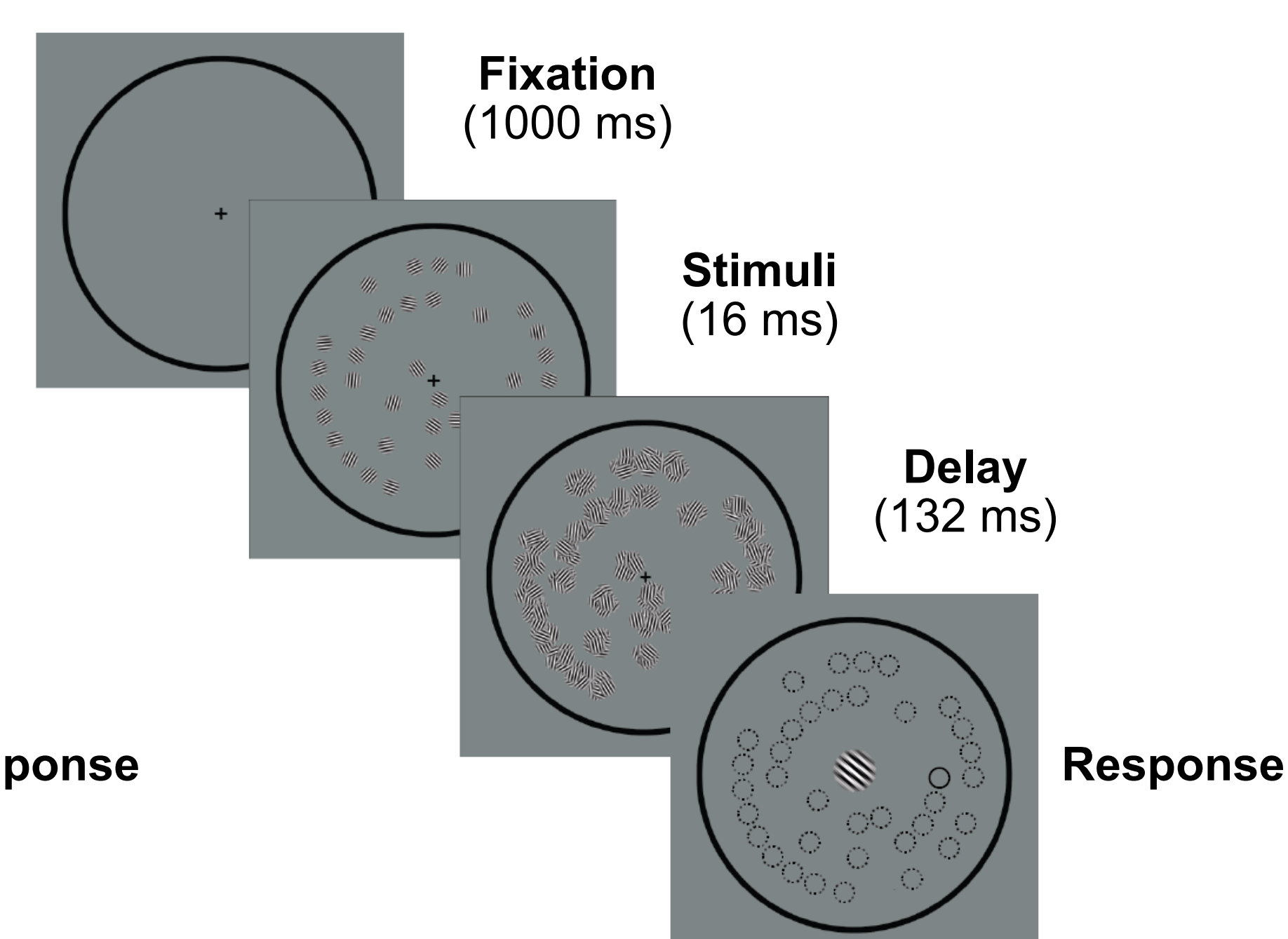


Experiment 1 task and trial sequence

1 item, 1000 ms trials

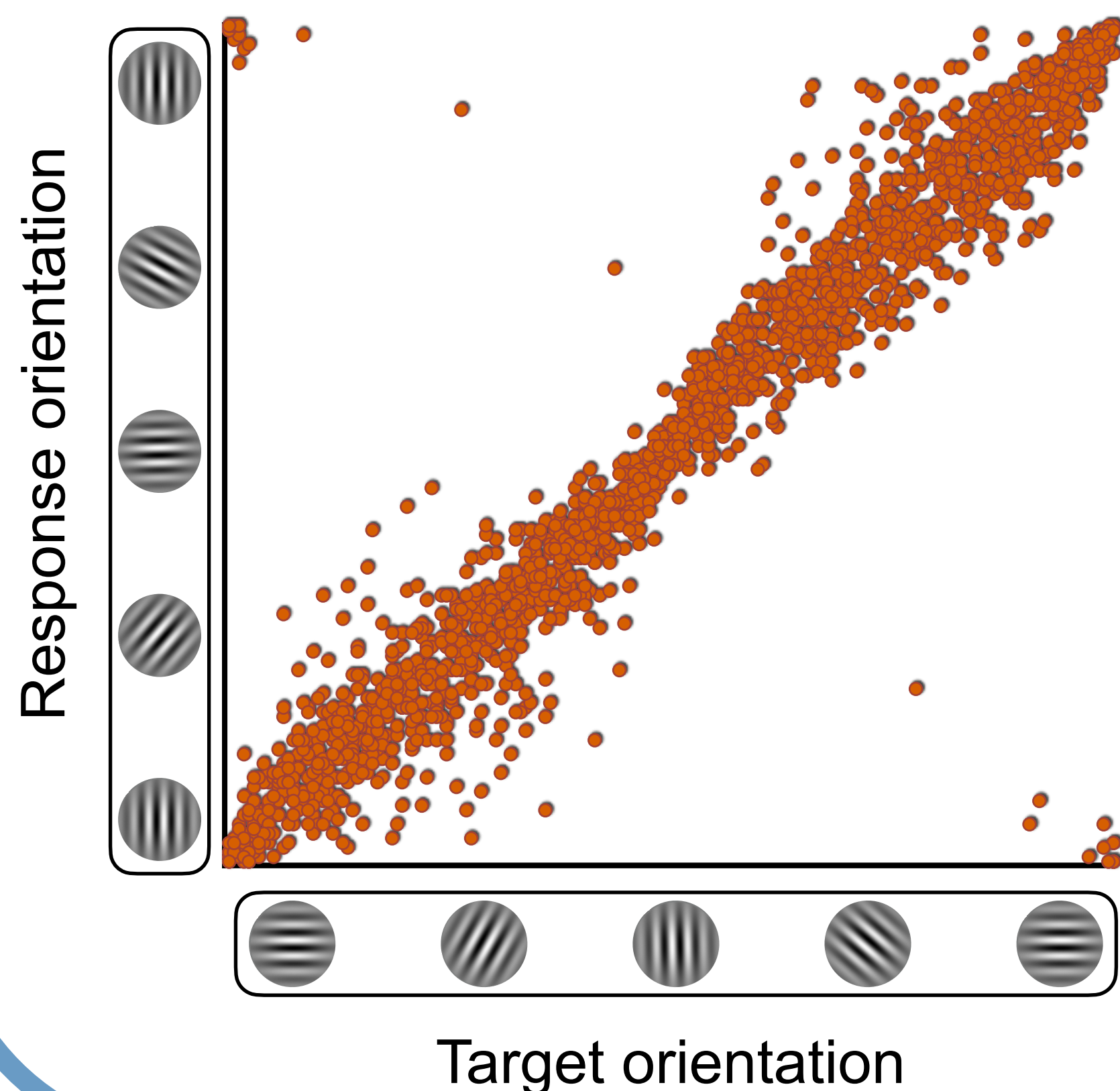


36 item, 16 ms trials

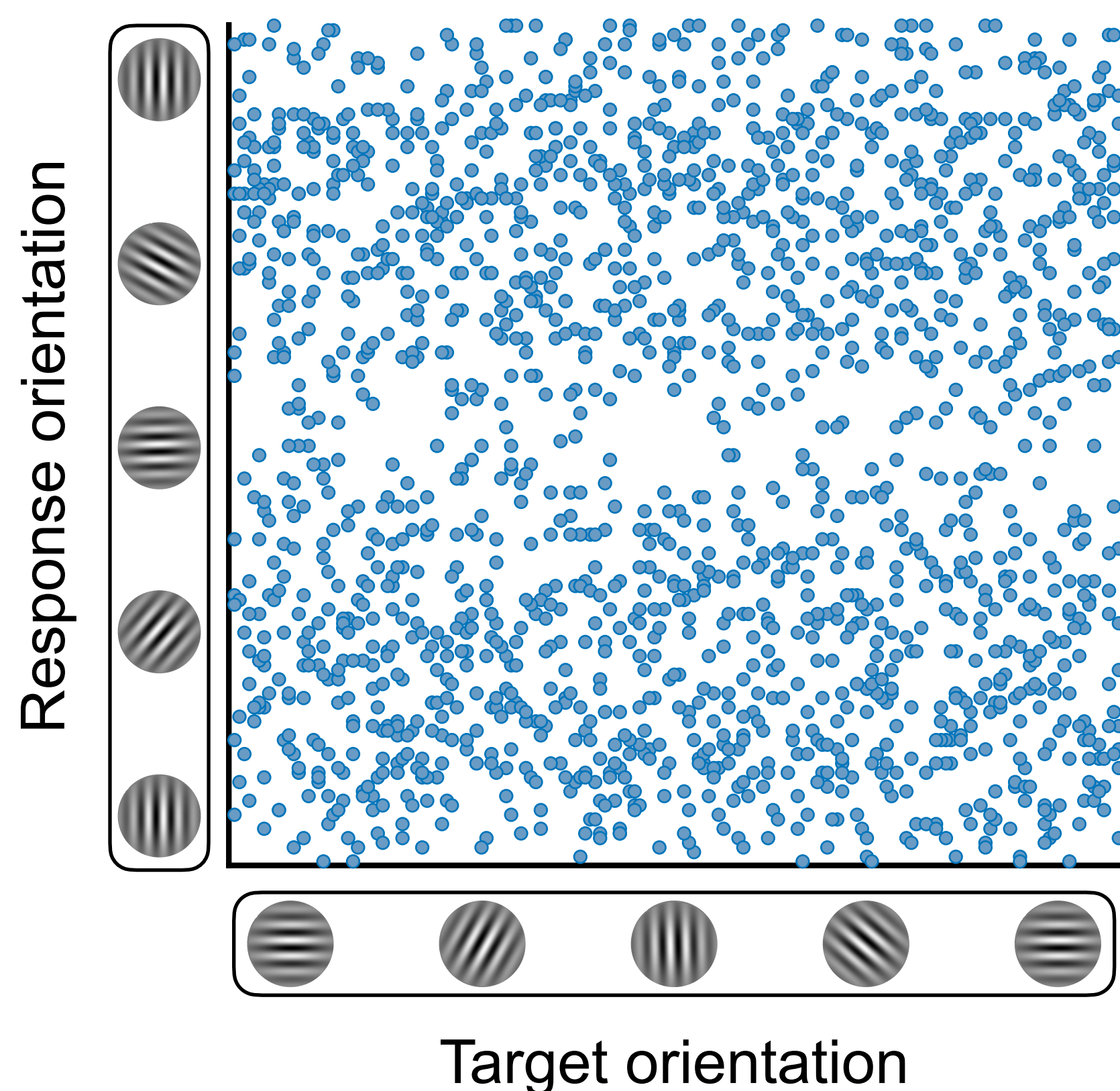


Experiment 1 results

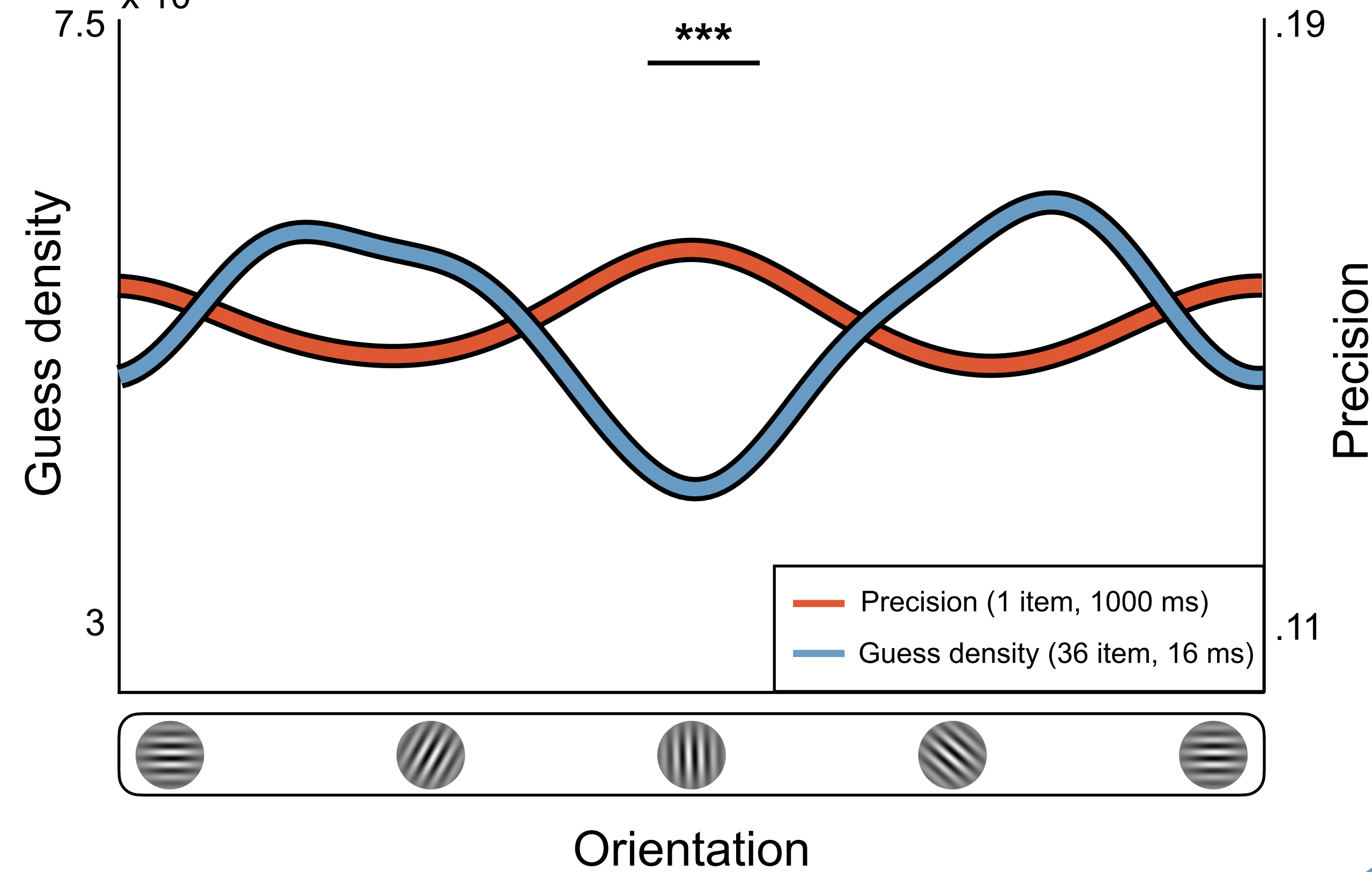
1 item, 1000 ms trials



36 item, 16 ms trials

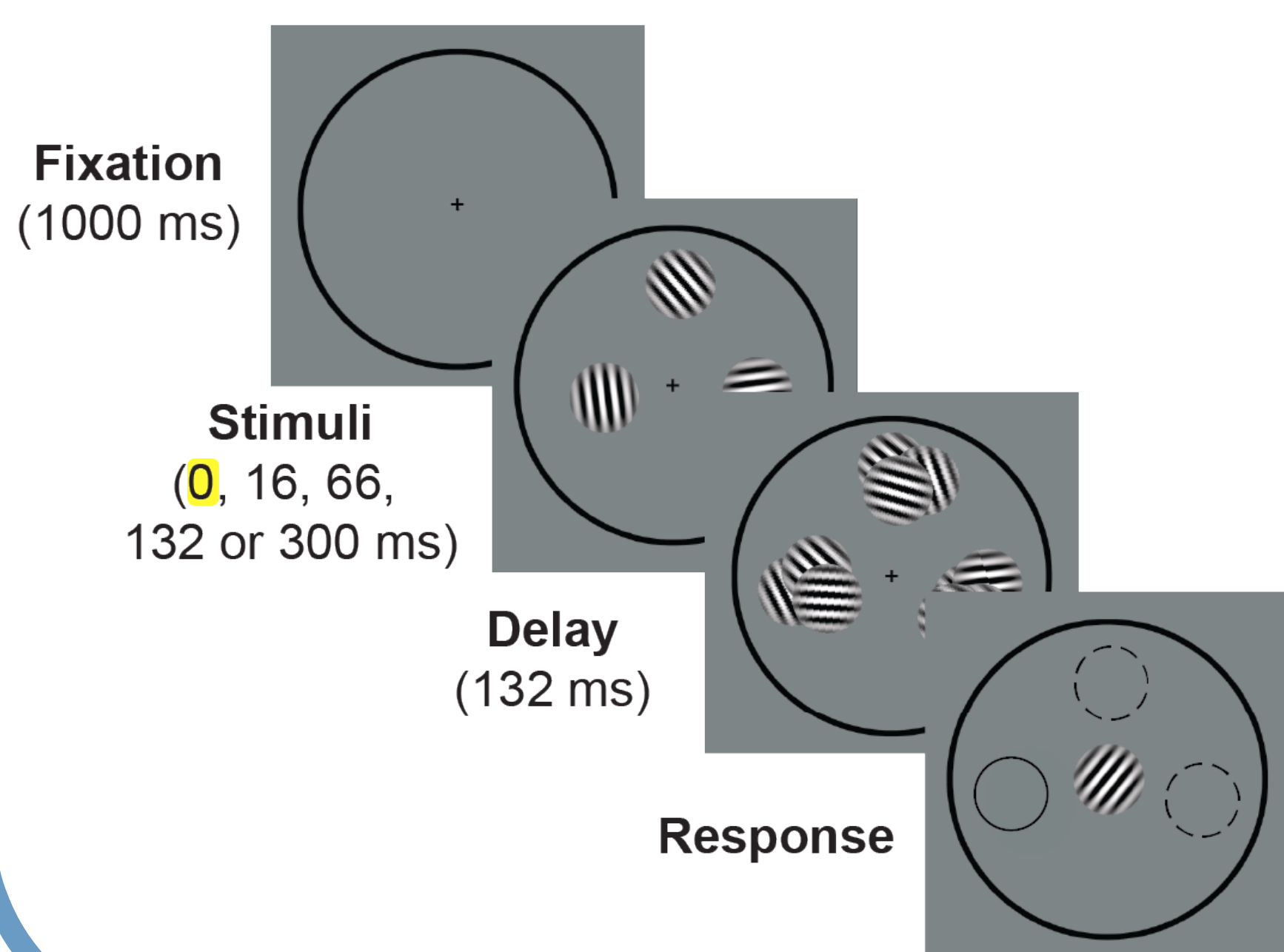


Guess density vs. precision



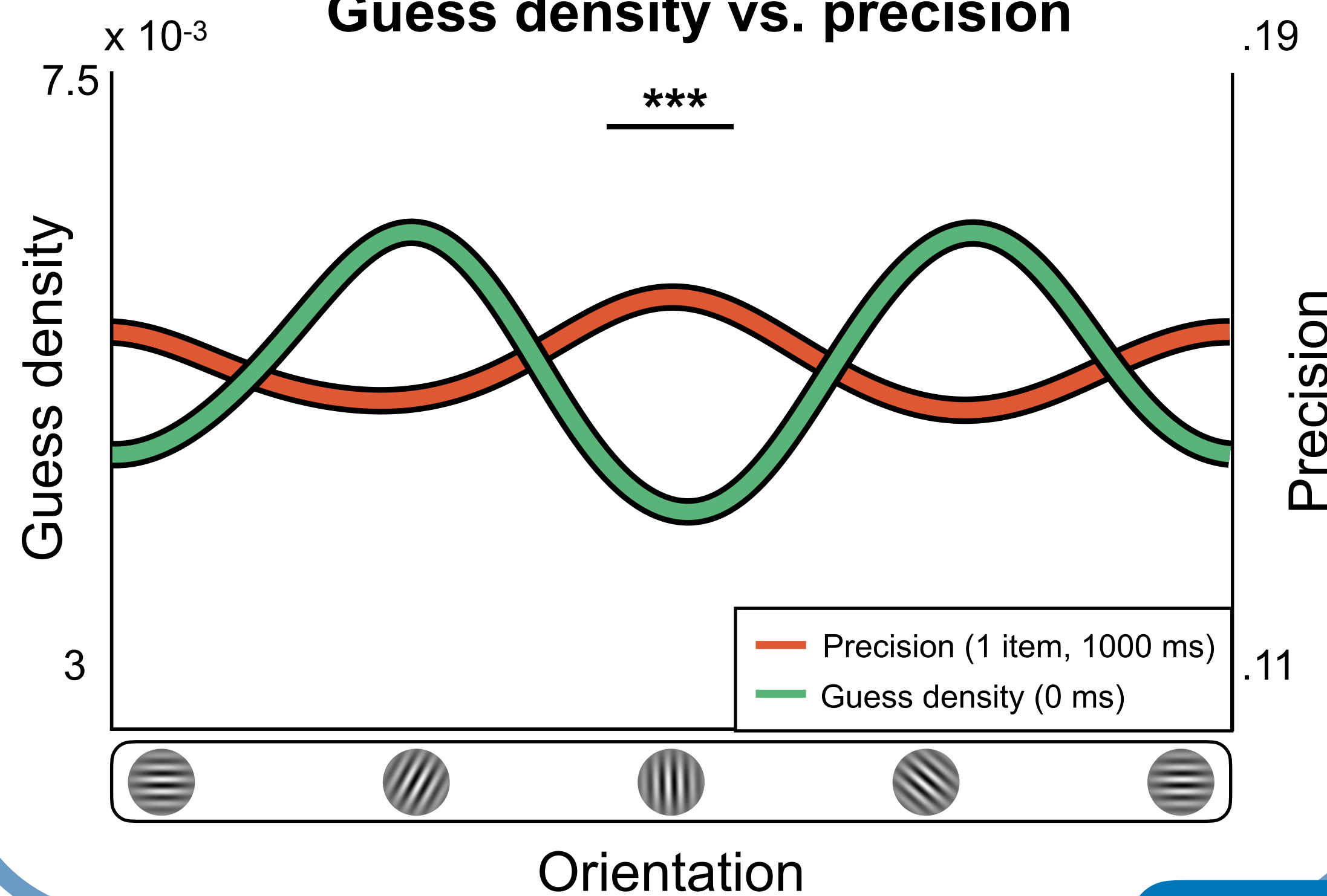
Experiment 2 task

Embedded stimulus-absent trials



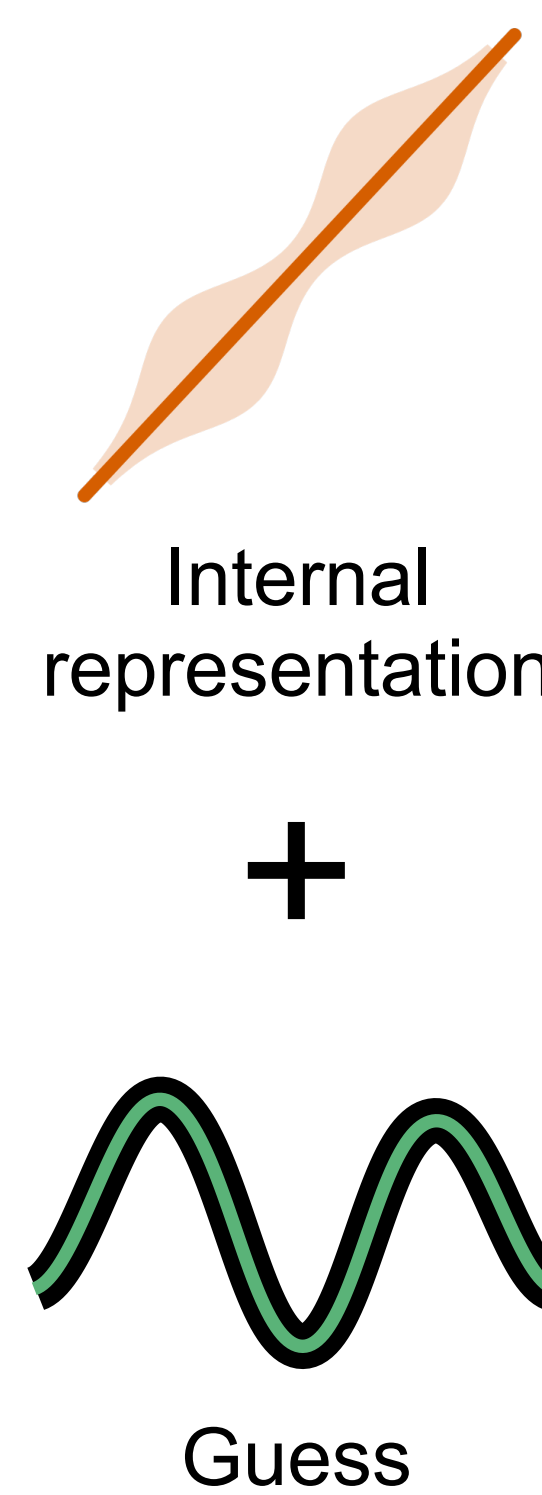
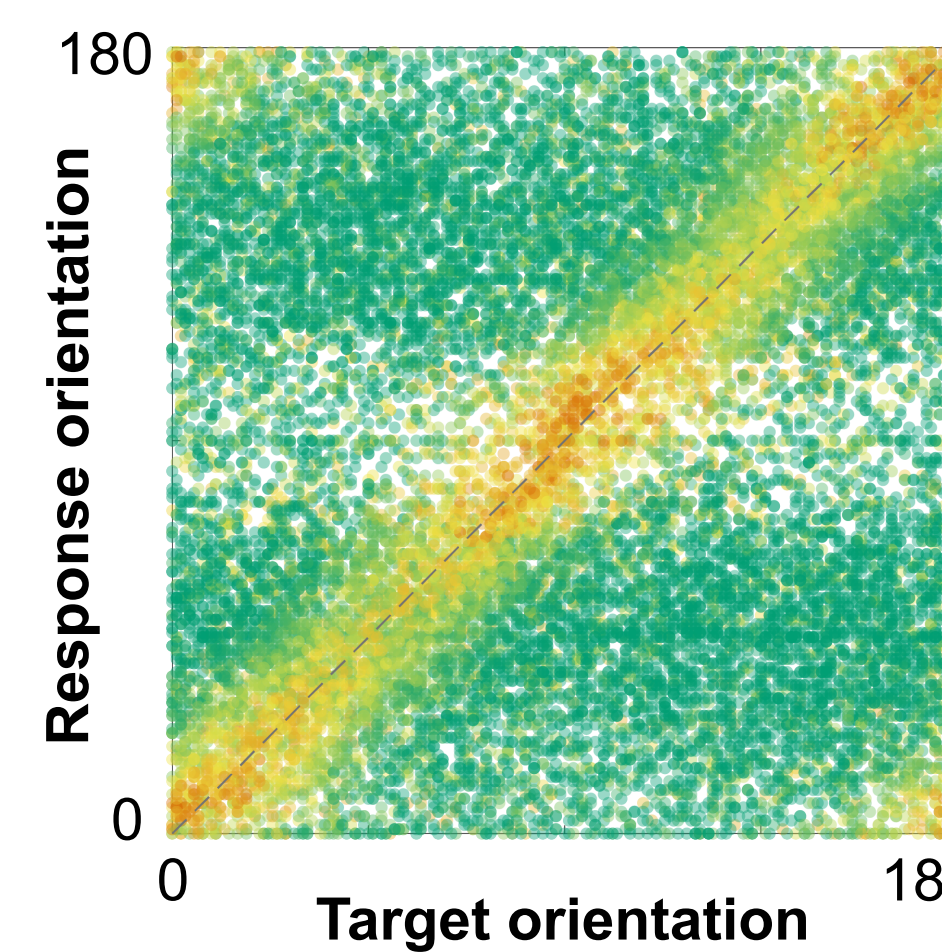
Experiment 2 results

Guess density vs. precision



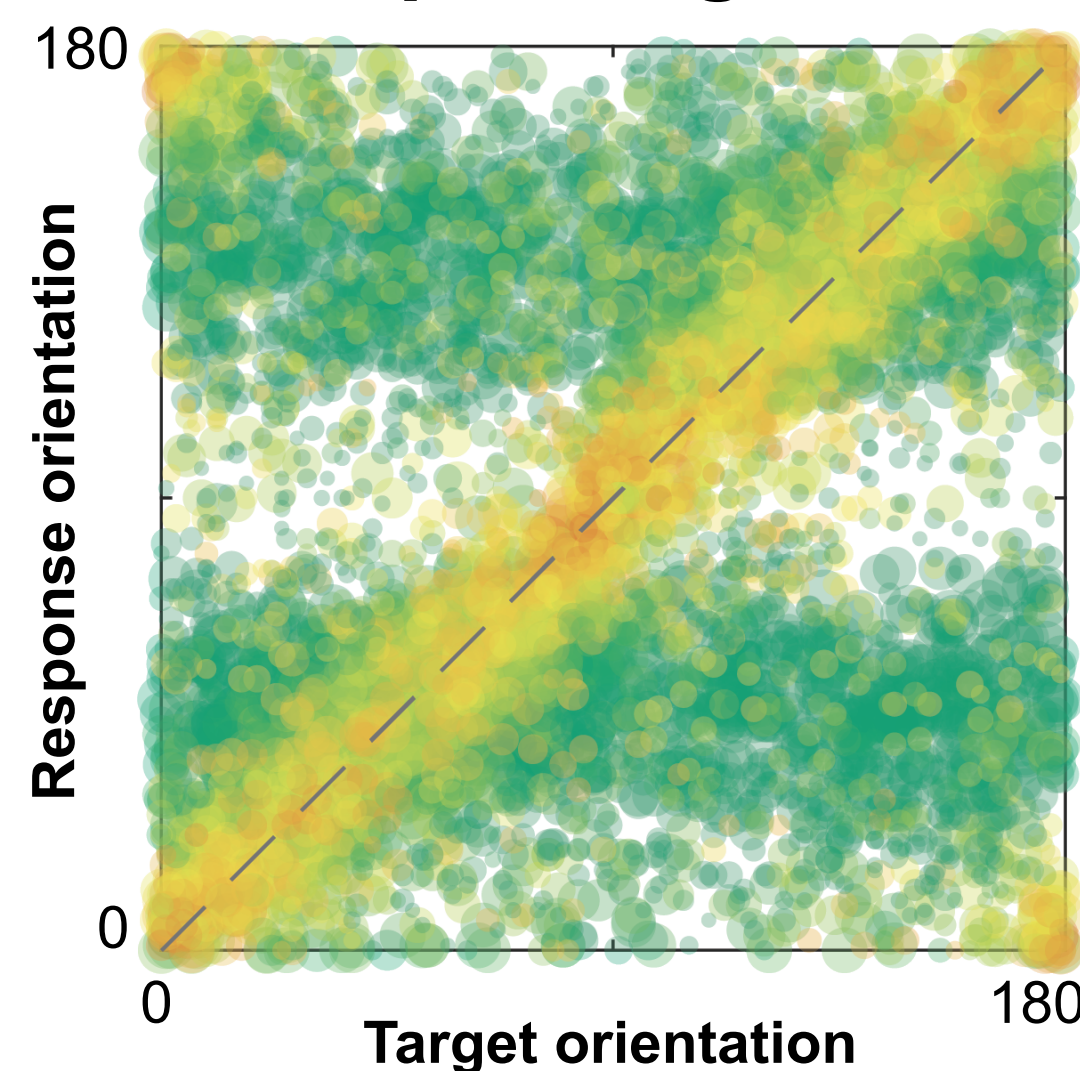
Mixture model

All trials

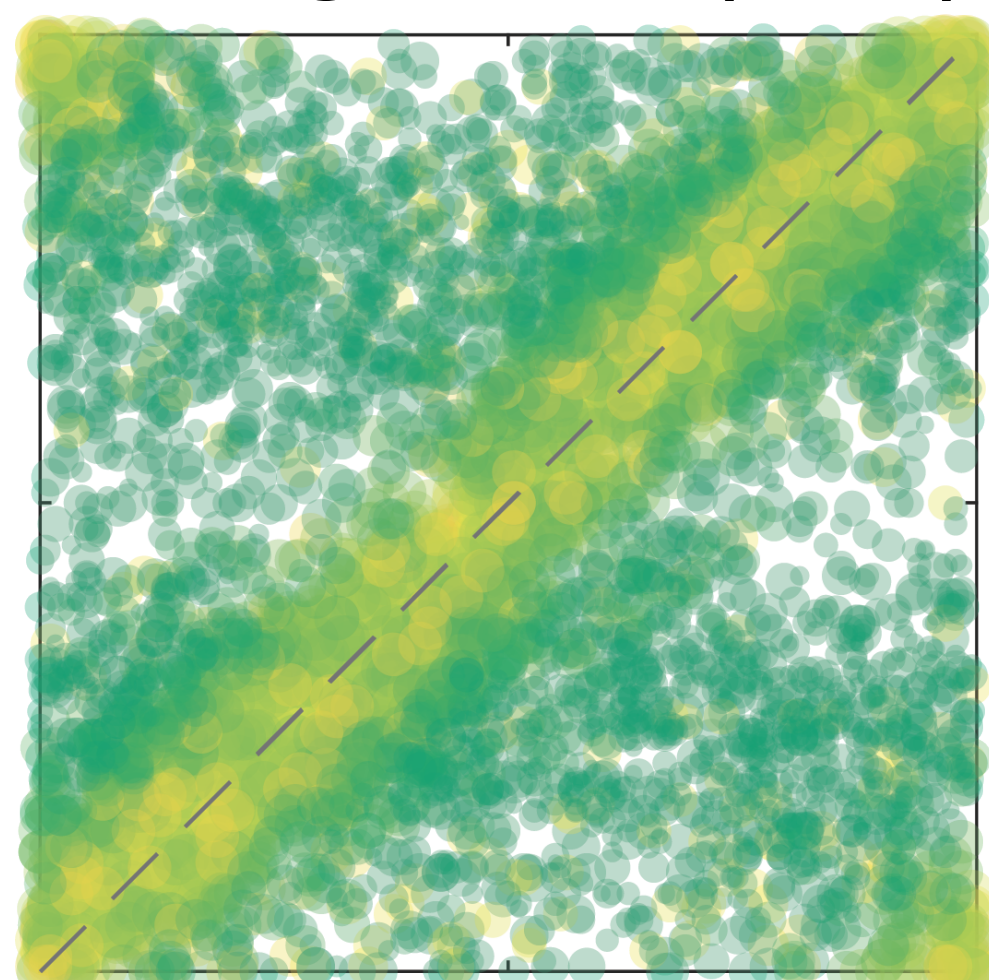


Model comparison

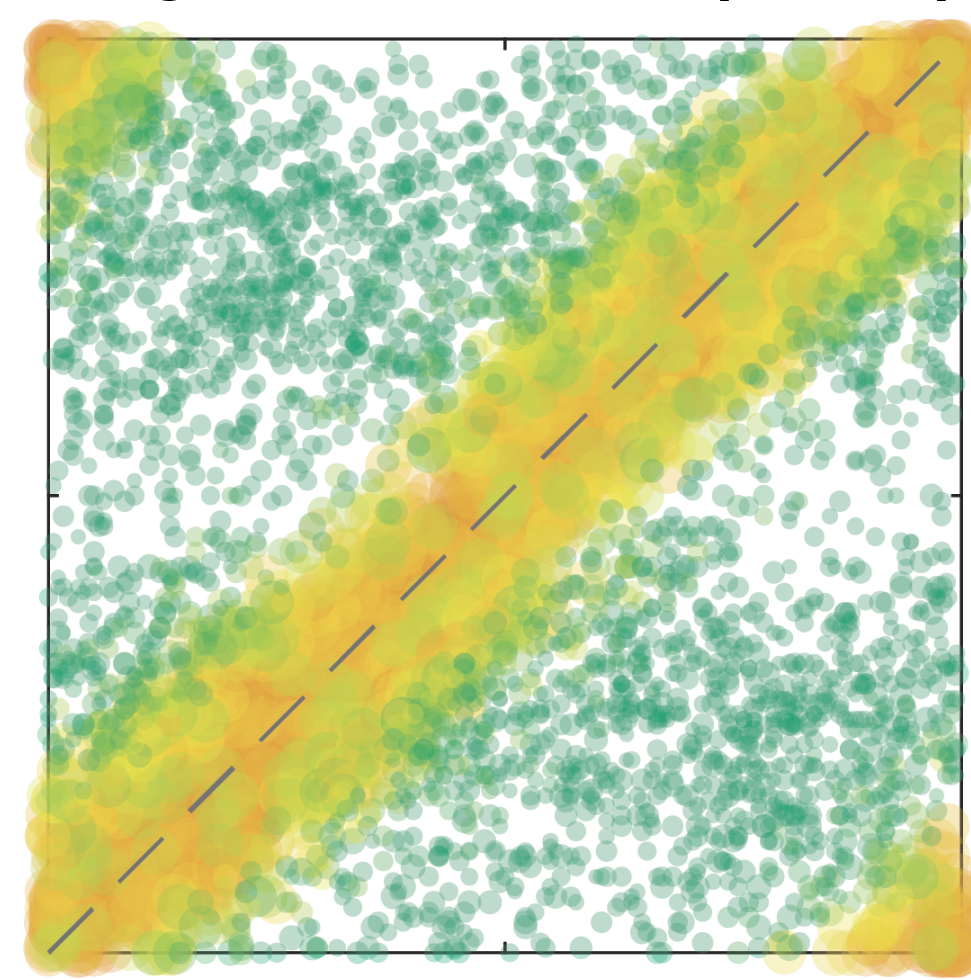
Empirical guess



Zhang & Luck (2008)

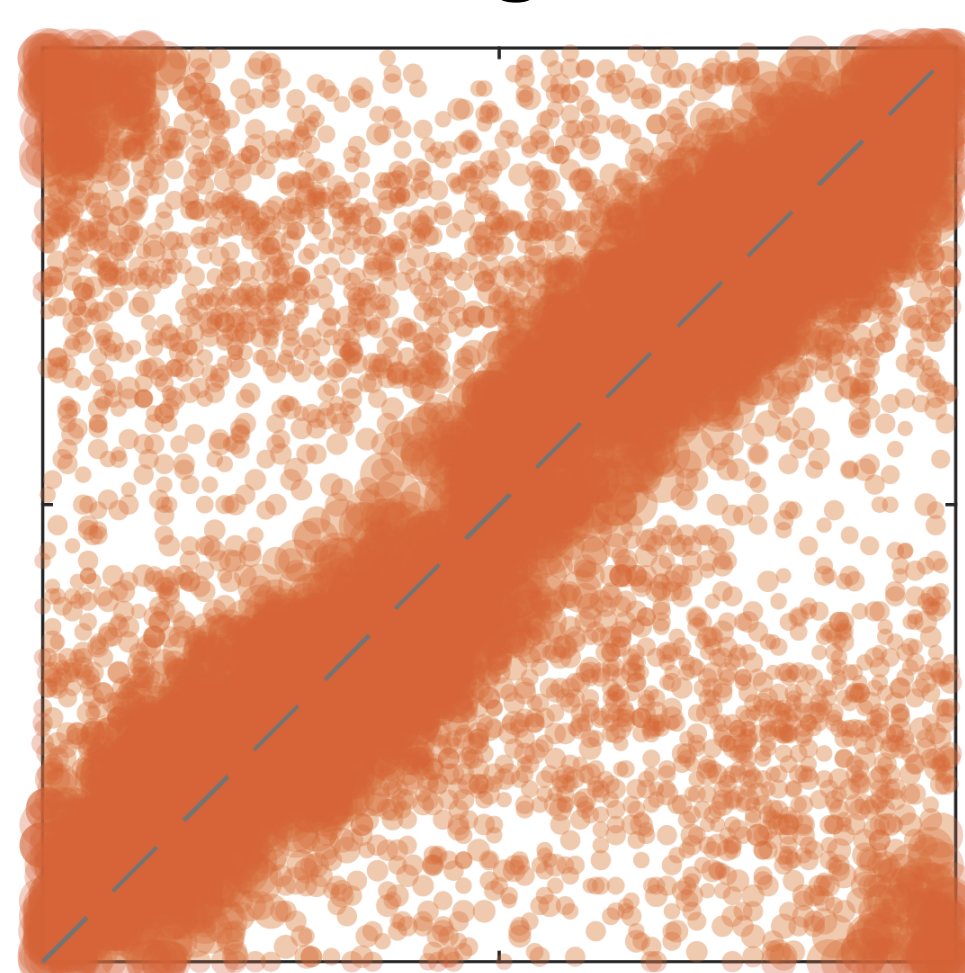


Bays & Husain (2008)

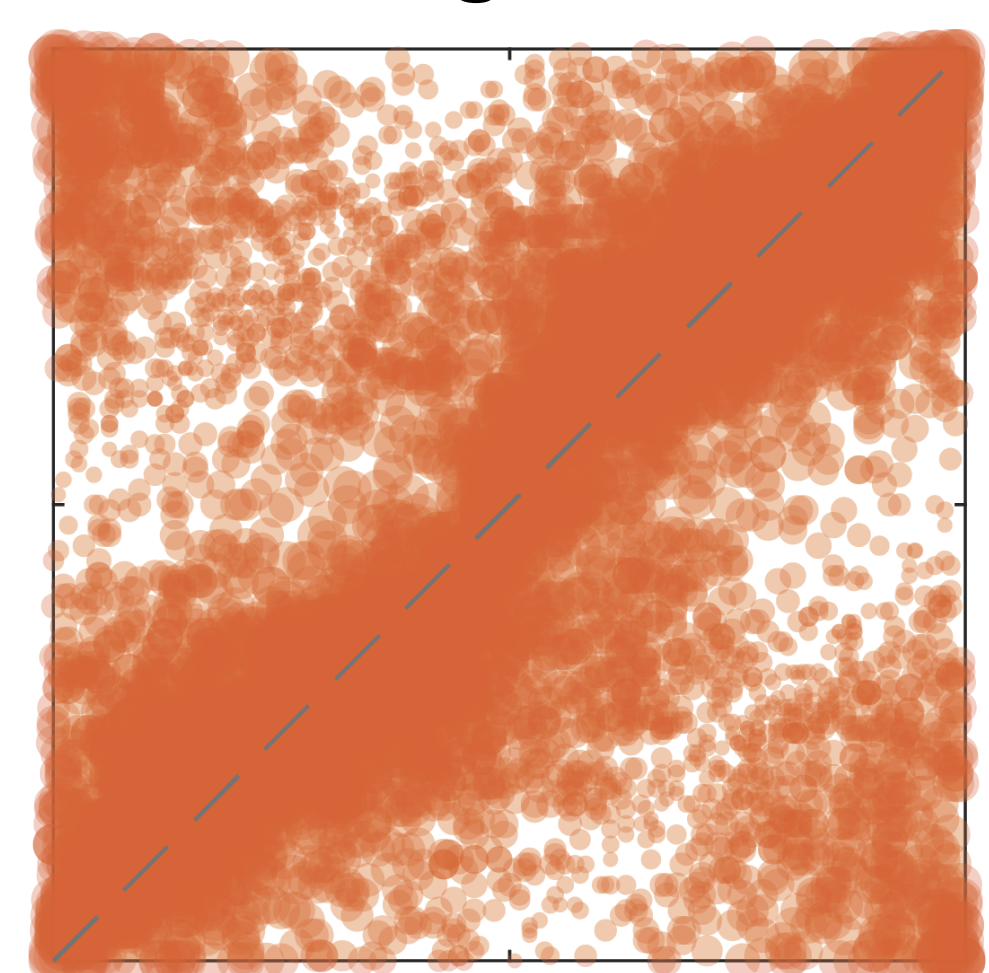


Posterior p(internal)

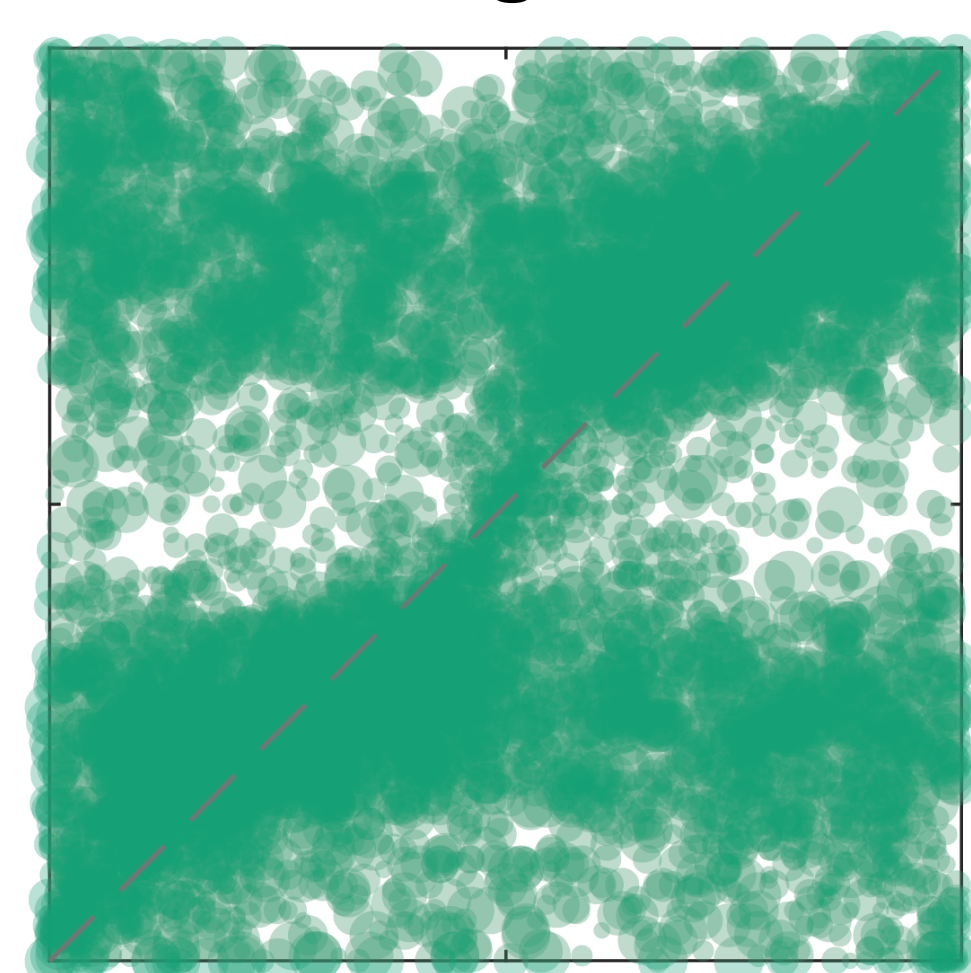
Van den Berg et al., 2012



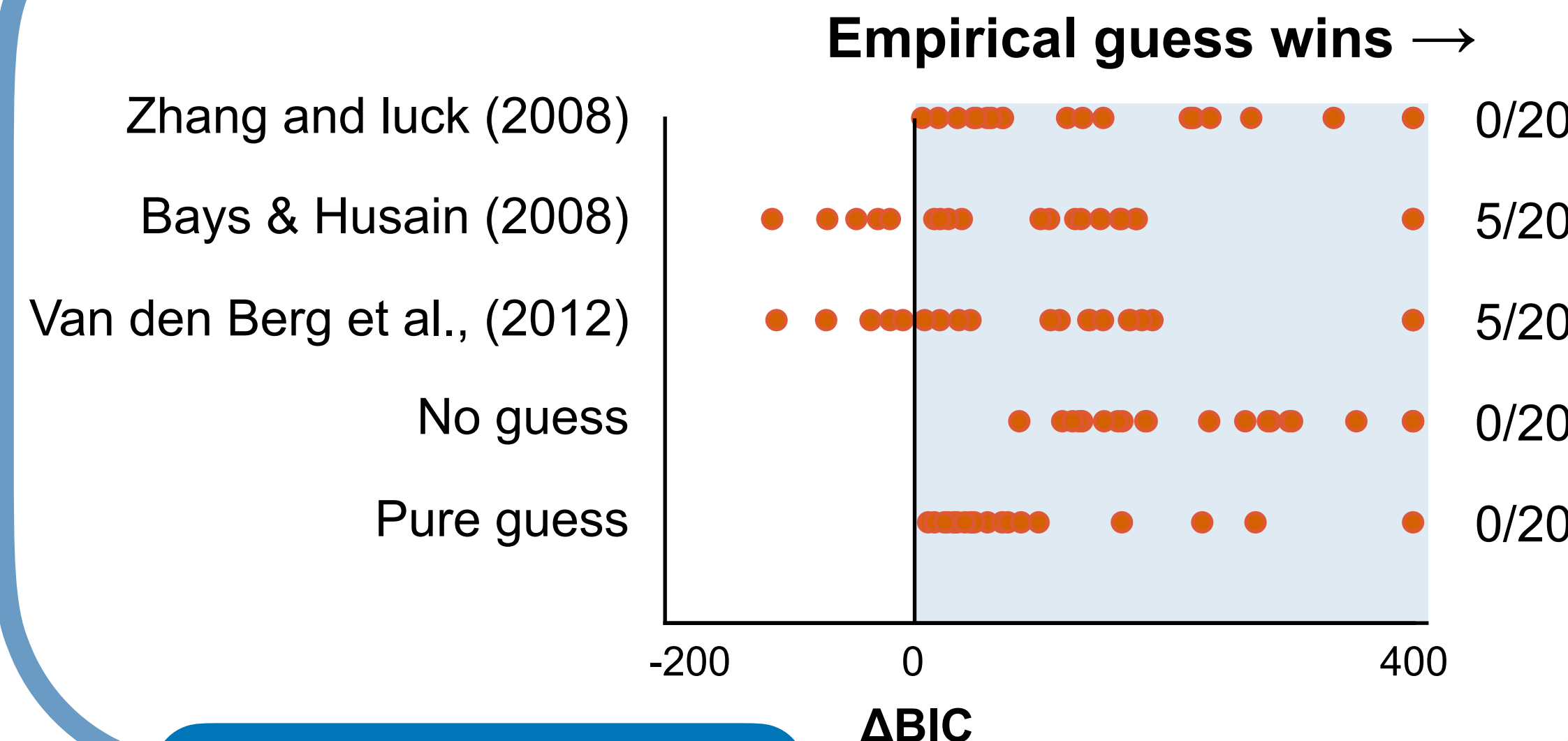
No guess



Pure guess



Empirical guessing Improves model fit



Conclusions

Models accounting for structured guessing **outperform** classic slot and resource accounts

Guessing is **structured** and **non-uniform**, contra lapse parameter assumptions